

Keeping the head cool

CPUs, often called the brains of a PC can heat up to the point where they can boil water (~100 degrees). But it is nearly guaranteed that they will kill themselves at that temperature by burning out and if the thermal sensing units are working well, the machine will halt well before you bring in eggs to boil over your CPU



Example of a liquid cooling heatsink

(strongly, not-recommended). To counter that heat, you will have to get a great heatsink with large surface area and a large fan to blow the heat away. If you are going to tax your CPU with lots of work for long hours (like rendering a 3D scene, video editing, audio format conversions or as a server that stays busy), we recommend you get yourself a liquid cooled heatsink. A liquid cool-

ing solution works better than simply installing fans because the specific heat capacity (rate at which heat is transferred) of water is about 4 times more than that of air. Also, water can be easily channeled in and out of the system while air turbulence caused by different moving fans can obstruct the heat flow and end up causing more trouble.

tioned. Normally, if the number of processing cores is less than the number of threads it can run simultaneously (it is clearly mentioned on the spec sheet), the chip comes with SMT (Simultaneous Multi-Threading) and you should be able to get more juice out of the processor than if there were none.

Also, you do not want to make a monster which overheats even when doing the minutest of tasks (like web browsing) and set the fans whirring. Power optimizations are just as important for that reason.

Many users do not take ‘security’ into consideration many-a-times. It is important to remember that if your PC is not secure, it can be very dangerous. The WannaCry ransomware which was recently in news only reinforces the importance of security. Though the mentioned ransomware was not because of lack of any hardware features, it is always good to look out of some extra protection. Higher end CPUs from Intel come with built in protection from multiple attacks that can be launched against your system right from the boot-up to when your system is properly working. **d**